The Murty-Simon bound at order 29

Reviewer-v4 hand-reduced candidate proof

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Murty-Simon at n=29: reviewer-v4 candidate proof

11 September 2026. Research directed by Paul Lenz; mathematical development, implementation and internal checking by ChatGPT/Geeps.

Status: serious candidate proof. Independent mathematical review, novelty assessment and independent computational reproduction remain OPEN. Reviewer-v4 replaces the proof-critical Delta=16 computation of reviewer-v3 by a hand threshold-tail argument. Historical editions, computations, bugs and audits remain preserved unchanged.

1. Candidate statement

Let G be a simple diameter-two edge-critical graph on 29 vertices. The candidate statement is

$$e(G) \leq 210 = \left\lfloor \frac{29^2}{4} \right\rfloor,$$

with equality if and only if

$$G \cong K_{14,15}.$$

A bipartite graph of diameter two is complete bipartite, so the bipartite case is immediate. For the non-bipartite dense case, the published Dailly-Foucaud-Hansberg dominating-edge theorem gives at most 208 edges when a dominating edge exists (apart from its order-six exception). Hence any counterexample at 210 or more edges has no dominating edge. A universal vertex gives a star and is sparse.

2. Delta=15 and the equality graph

A direct witness is an edge uv whose endpoints have no common neighbour. A two-step witness is a nonedge uv whose endpoints have exactly one common neighbour. Every critical edge is covered by one such witness. A direct witness covers one edge; a two-step witness covers at most the two edges of its unique length-two path.

Because there is no dominating edge, every witness pair uv satisfies

$$d(u) + d(v) \leq 28. \tag{2.1}$$

Assume Delta=15. Put

$$\varepsilon_x = 15 - d(x), \qquad L = \{x : \varepsilon_x \geq 2\}, \qquad O = \{x : \varepsilon_x = 1\}.$$

Let h=|L|, o=|O|, and

$$T = 29 \cdot 15 - 2e(G).$$

Every witness has deficit sum at least two, so

$$2h + o \leq T. \tag{2.2}$$

Count all edges incident with L directly. An edge entirely outside L is witnessed either by an O-O witness, of capacity at most two, or by a missing L-X pair, of capacity at most one. Hence

$$e(G) \leq \binom{h}{2} + h(29 - h) + o(o - 1). \tag{2.3}$$

At e(G)=211, T=13. The maximum right sides of (2.3) for h=0,...,6 are

`156, 138, 127, 123, 126, 136, 153,`

all below 211. Thus Delta=15 is impossible at 211 and therefore at every larger edge count.

At e(G)=210, T=15. The corresponding maxima are

`210, 184, 165, 153, 148, 150, 159, 175.`

Equality forces h=0 and o=15. All witnesses then lie inside O. Separating direct O-edges from two-step O-nonedges,

$$210 \leq e(G[O]) + 2\left(\binom{15}{2} - e(G[O])\right) = 210 - e(G[O]),$$

so O is independent. Its 15 vertices each have degree 14 and therefore meet all 14 vertices outside O. These 210 cross edges exhaust the graph. Hence

$$G = K_{15,14}. \tag{2.4}$$

3. The Delta=16 bridge

Put $H = \overline{G}$, choose a minimum-degree vertex v of H, and set

```
A=N_H(v),          |A|=a=12,
B=V(H)\N_H[v],     |B|=b=16.
```

Let $C = H[A]$, and let F be the complement of C on A; write $d_i = d_F(i)$. For every missing unordered B-pair choose exactly one cross quasi-edge representative `ui->w`, with exception w in B. Call these cross-edges selected and all other existing A-B edges residual. For $i \in A$, let R_i be its residual degree into B and set

$$s_i = \max(0, d_i - R_i), \qquad S = \sum_i s_i.$$

For $u \in B$ let ρ_u be its residual A-degree and put $r = \sum_u \rho_u$. Finally set

$$t = e(G) - 208.$$

The self-contained graph-to-model bridge is preserved at

`project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md`.

Reviewer-v4 uses only the following consequences of that bridge:

1. **Exact ledger and demand:**
$$S \geq r + 2t. \tag{3.1}$$
2. **Residual activity:** if t>0, then
$$\rho_u \geq 1 \qquad (u \in B). \tag{3.2}$$
3. **Selected-incidence forcing:** at every selected incidence carrying label i,
$$s_i \leq \rho_u. \tag{3.3}$$
4. **Selected-source capacity:** a selected source has at least one selected edge, so
$$\rho_u \leq a - 1 = 11. \tag{3.4}$$
Consequently every positive demand satisfies 1<=s_i<=11.
5. **Threshold capacity:** for h>=2, with
$$W_h = \sum_{i: s_i \geq h} s_i, \qquad z_h = |\{u \in B : \rho_u \geq h\}|,$$
one has, whenever W_h>0,
$$z_h \geq h, \qquad 2W_h \leq z_h^2 - z_h + h(h+1). \tag{3.5}$$

The threshold-capacity proof had a non-blocking sign/order typo in an earlier explanatory sentence; the corrected identity is recorded in the historical lemma and is the direction used in (3.5). No numerical result changed.

Crucially, reviewer-v4 does **not** use the charging inequality, isolated-C lemma, residual-row enumeration, grouped LP or Farkas certificates in the Delta=16 branch.

4. Convert threshold capacity to a demand-only inequality

For h>=2 define

$$g_h(W) = \begin{cases} 0, & W = 0, \\ \min\{z \in \{h, \dots, 16\} : 2W \le z^2 - z + h(h+1)\}, & W > 0. \end{cases} \tag{4.1}$$

If the set in (4.1) is empty, the demand vector is already impossible. Otherwise (3.5) gives z_h>=g_h(W_h).

Because every rho_u>=1 when t>0 and rho_u<=12,

$$r = 16 + \sum_{h=2}^{12} z_h \ge 16 + \sum_{h=2}^{11} g_h(W_h). \tag{4.2}$$

Combining (3.1) and (4.2), every positive-surplus Delta=16 graph satisfies

$$Q(s) := S - \sum_{h=2}^{11} g_h(W_h) \ge 16 + 2t. \tag{4.3}$$

The residual degree sequence has disappeared.

5. Hand threshold-tail lemma: Q(s)<=18

Let

$$N_h = |\{i : s_i \ge h\}|, \qquad p = N_1, \qquad d_h = N_h - g_h(W_h),$$

and

$$D(s) = \sum_{h=2}^{11} d_h.$$

The Ferrers-tail identity gives

$$S = p + \sum_{h=2}^{11} N_h,$$

and hence

$$Q(s) = p + D(s). \tag{5.1}$$

Since p<=12, it suffices to prove D(s)<=6. Write

$$C_h(z) = \frac{z(z-1) + h(h+1)}{2}.$$

5.1 Levels h>=7 are nonpositive

Fix h>=7 and N=N_h. If N=0 then d_h=0; if 0<N=h>N and d_h<0. If N>=h, W_h>=hN. Were g_h<=N-1, then hN<=C_h(N-1). Write N=h+e. Since N<=12,

$$0 \le e \le 12 - h \le 5.$$

Twice the difference is

$$2hN - [(N-1)(N-2) + h(h+1)] = -e^2 + 3e + 2h - 2. \tag{5.2}$$

This concave quadratic is positive at both endpoints e=0 and e=5 (the latter gives at least 2), contradiction. Thus g_h>=N_h and

$$d_h \le 0 \qquad (h \ge 7). \tag{5.3}$$

Therefore clipping every demand above 6 down to 6 cannot decrease D: for h<=6, N_h is unchanged and W_h decreases; for h>=7, the new deficit is zero while the old one was nonpositive.

5.2 Clip 6 to 5

Assume max s_i<=6 and let k=N_6. For k<=10, g_6(6k)>=k, hence d_6<=0. For k<6 this is immediate from g_6>=6; for 6<=k<=10, testing z=k-1 gives

$$12k - [(k-1)(k-2) + 42] = -e^2 + 3e + 10 > 0,$$

where e=k-6 in {0,1,2,3,4}.

Only k=11,12 can have positive d_6, and in both cases d_6=1. If k=11, let l be 1 when the remaining demand is five and 0 otherwise. At h=5,

```
before: W_5=66+5l,
after:  W'_5=55+5l.
```

Using C_5(10)=60, C_5(11)=70, C_5(12)=81, g_5 drops by at least one. If k=12, W_5 drops from 72 to 60 and g_5 drops from 12 to 10. Thus clipping 6->5 cannot decrease D.

5.3 Clip 5 to 4

Assume max s_i<=5 and let k=N_5. For k<=9, d_5<=0; for 5<=k<=9 this follows from

$$10k - [(k-1)(k-2) + 30] = -e^2 + 3e + 8 > 0,$$

with e=k-5 in {0,1,2,3,4}.

For k=10,11, d_5=1. Let l be the number of fours among the remaining entries. At h=4 the complete possibilities are

```
(k,l)   W_4 -> W'_4     g_4 -> g'_4
(10,0)   50 -> 40        10 -> 9
(10,1)   54 -> 44        10 -> 9
(10,2)   58 -> 48        11 ->10
(11,0)   55 -> 44        10 -> 9
(11,1)   59 -> 48        11 ->10
```

so the lower-level gain pays for d_5. If k=12, the vector is (5^12), for which directly

$$D(5^{12}) = (12-12) + (12-11) + (12-11) + (12-10) = 4 \leq 6.$$

Thus every remaining vector either already satisfies the target or can be clipped 5->4 without decreasing D.

5.4 Clip 4 to 3

Assume max s_i<=4 and let k=N_4. For k<=8, d_4<=0; for 4<=k<=8 this follows from

$$8k - [(k-1)(k-2) + 20] = -e^2 + 3e + 6 > 0,$$

with e=k-4 in {0,1,2,3,4}.

For k=9,10, d_4=1; checking the at most four possible numbers l of remaining threes shows g_3 drops by at least one under 4->3. For k=11, d_4=2: if the remaining entry is at most two, g_3 drops by two; if it is three, the change (3,4$^{11)\text{->}(3}$12) drops g_3 by one and g_2 by one. For k=12, (4$^{12)\text{->}(3}$12) likewise drops g_3 and g_2 by one each, paying for d_4=2.

Hence it remains only to treat demands in {0,1,2,3}.

5.5 Final two-level inequality

Put

$$x = N_2, \qquad y = N_3, \qquad 0 \leq y \leq x \leq 12.$$

Then

$$W_2 = 2x + y, \qquad W_3 = 3y,$$

and

$$D = x + y - g_2(2x + y) - g_3(3y). \tag{5.4}$$

If 0<=y<=6, then

$$x - g_2(2x + y) \leq 5, \qquad y - g_3(3y) \leq 0,$$

so D<=5. For x>=8, the first inequality follows because capacity at z=x-6 is too small:

$$2x - C_2(x-6) = \frac{-x^2 + 17x - 48}{2} > 0;$$

the smaller x cases follow from g_2>=2. The y inequality is immediate for y<=3 and for 4<=y<=6 follows from

$$3y - C_3(y-1) = \frac{-y^2 + 9y - 14}{2} > 0.$$

If 7<=y<=10, then

$$y - g_3(3y) \leq 2$$

because

$$3y - C_3(y-3) = \frac{-y^2 + 13y - 24}{2} > 0.$$

Also

$$x - g_2(2x + y) \leq 4,$$

because capacity at z=x-5 is too small:

$$(2x + y) - C_2(x-5) = \frac{-x^2 + 15x + 2y - 36}{2} \geq \frac{-x^2 + 15x - 22}{2} > 0$$

for 7<=x<=12. Hence D<=6.

Finally, if 11<=y<=12, then

$$y - g_3(3y) \leq 3$$

because

$$3y - C_3(y-4) = \frac{-y^2 + 15y - 32}{2} > 0.$$

Since y<=x<=12, x is 11 or 12, and

$$x - g_2(2x + y) \leq 3$$

because

$$(2x+y) - C_2(x-4) = \frac{-x^2+13x+2y-26}{2} \geq \frac{-x^2+13x-4}{2} > 0.$$

Thus D<=6 in all cases. From (5.1),

$$\boxed{Q(s) \leq 18}. \tag{5.5}$$

A small exact integer checker independently regression-tests the clipping endpoint tables and the final 91 (x,y) pairs, while a separate historical checker exhausts all 1,352,078 nondecreasing 12-demand multisets. Both agree with (5.5), but neither computation is a logical premise of reviewer-v4.

6. Consequence: all dense Delta=16 cases disappear

From (4.3) and (5.5),

$$16 + 2t \leq 18,$$

so

$$t \leq 1. \tag{6.1}$$

But e(G)>=210 means t=e(G)-208>=2. Therefore

$$\boxed{\Delta = 16 \text{ is impossible for every } e(G) \geq 210.} \tag{6.2}$$

This single hand argument replaces the entire proof-critical Delta=16 finite stack in reviewer-v3: demand enumeration, residual-row enumeration, Hall pruning, supplement refinement, corrected grouped LP and exact Farkas certificates. Those artifacts remain preserved as independent computational corroboration and audit history.

7. Delta=17

For Delta=17, a=11. The bridge charging inequality requires

$$17 + 2t \leq \sum_{i=1}^{11} \frac{s_i(12-2s_i)}{11-s_i}.$$

For every integer 0<=s<=10,

$$\frac{s(12-2s)}{11-s} \leq \frac{16}{7},$$

since

$$\frac{16}{7} - \frac{s(12-2s)}{11-s} = \frac{2(s-4)(7s-22)}{7(11-s)} \geq 0.$$

Thus the right side is at most 176/7<29. At 210 edges the required left side is already 29 and increases with e(G). Hence Delta=17 is impossible throughout the relevant range.

8. Delta=18 through 27

Let h be the largest integer for which at least h residual rows have degree at least h. A label of demand s_i needs s_i distinct selected sources of residual degree at least s_i, so s_i<=h and S<=ah. Residual activity gives

$$r \geq h^2 + (b-h) = b + h(h-1).$$

Combining with S>=r+2t,

$$b + 2t \leq (a+1)h - h^2 \leq \left\lfloor \frac{(a+1)^2}{4} \right\rfloor = \left\lfloor \frac{(29-b)^2}{4} \right\rfloor. \tag{8.1}$$

Every b=18,...,27 violates (8.1) already at 210 edges, and increasing e(G) only strengthens the contradiction. Delta=28 is the universal-vertex/star case.

9. Assembly

At e(G)>=211:

- Delta<=14 is impossible by the degree sum;
- Delta=15 is impossible by Section 2;
- Delta=16 is impossible by Section 6;
- Delta=17 is impossible by Section 7;
- Delta=18,...,27 are impossible by Section 8;
- Delta=28 gives a star.

Hence e(G)<=210.

At e(G)=210, Delta<=14 is again impossible by degree sum. Sections 6-8 exclude every Delta>=16. Therefore Delta=15, and Section 2 forces

$$G \cong K_{14,15}.$$

This establishes the stated **candidate** order-29 result, subject to the independent-review boundary below.

10. What reviewer-v4 no longer depends on

For the theorem proof, reviewer-v4 does not require:

- Fan's 1987 density theorem;
- the N29 minimal trusted computational kernel;
- the corrected late cumulative-threshold LP;
- exact Farkas certificates;
- residual-row enumeration;
- source-capacity Hall pruning;
- the isolated-C lemma in the Delta=16 branch;
- the Delta=16 charging enumeration;
- the historical upper-range Delta=16 scans;
- the active RX-Hall / 3-D potential programme.

All remain preserved where historically relevant.

11. Current trust boundary

The highest-value independent mathematical targets are now substantially narrower:

1. the complement/quasi-edge selected/residual construction;
2. the exact ledger and demand implication S>=r+2t;
3. residual activity rho_u>=1 for t>0;
4. selected-incidence forcing s_i<=rho_u and selected-source capacity rho_u<=11;
5. the threshold-capacity lemma (3.5);
6. the hand threshold-tail clipping proof D<=6;
7. the Delta=15 witness-capacity/equality argument;
8. the Delta=17 pointwise charging bound and Delta>=18 residual h-index argument;
9. the cited Dailly-Foucaud-Hansberg dominating-edge theorem.

A counterexample to any universal bridge lemma overrides every downstream conclusion. No unrestricted Murty-Simon theorem, external endorsement or novelty determination is claimed.

12. Canonical supporting files

- Frozen self-contained bridge: `project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.`
- Detailed threshold-tail hand proof: `project/research/general_n/2026-09-11-threshold-tail-collaps`
- Threshold-tail audit README: `project/research/general_n/2026-09-11-threshold-tail-collapse-`
- Historical minimal-kernel report: `project/reviews/n29/2026-09-08-redteam-restart-v1/MINIMAL_KER`
- Blind hostile review follow-up: `project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/FOL`
- Cross-order bridge re-audit: `project/reviews/cross-cutting/2026-09-11-n29-cross-order-reaudit-`

Reviewer-v3 and all earlier packages remain frozen for provenance.

Appendix A. Frozen self-contained Delta=16 graph-to-model bridge

11 September 2026. Reviewer-v3 bridge. Research directed by Paul Lenz; mathematical development and checking by ChatGPT/Geeps. Hardened after a blind external-assistant red-team.

Status: candidate mathematics; independent expert review remains open. This version is deliberately self-contained at the main trust boundary. In particular, the threshold-capacity argument is proved here rather than merely cited from a companion note.

1. Setup

Let `G` be a simple diameter-two edge-critical graph on 29 vertices with `Delta(G)=16`. Put `H=complement(G)`. Choose a minimum-degree vertex `v` of `H`. Then

```
d_H(v)=29-1-16=12.
```

Set

```
A=N_H(v),              |A|=a=12,
B=V(H)\N_H[v],         |B|=b=16.
```

Let `C=H[A]`, and let `F` be the complement of `C` on A. Write `d_i=d_F(i)`.

For an edge count `m=e(G)`, define

```
t=m-b(29-b)=m-208.
```

The proof-critical dense scopes are `t=2,3,4` for `m=210,211,212`; the upper-range hand arguments below also use larger t.

2. Quasi-edges and one representative per missing unordered B-pair

Take a missing pair `uw` of `H[B]`. Adding `uw` to H corresponds to deleting the critical edge `uw` from G. In `H+uw` a new adjacent total-dominating pair appears.

That pair must use u or w because no other adjacency changed. It cannot be `{u,w}` because both still miss v. Hence, after interchanging u,w if necessary, there is an existing edge `ui` of H such that

```
N_H(u) union N_H(i) = V(H)\{w}.
```

The auxiliary i lies in A because the pair must dominate v. Write

```
ui -> w.
```

For every missing **unordered** pair `{u,w}` in `H[B]`, choose exactly one such cross-edge, after orienting the pair if necessary. Call it **selected**. Every other existing A-B edge is **residual**.

A selected edge determines its B-source u and its unique exception w. Therefore it recovers its indexing missing unordered B-pair. Consequently:

1. two different missing unordered B-pairs cannot choose the same selected cross-edge;
2. opposite orientations of one unordered pair cannot both be designated selected;
3. at a fixed source, selected labels and supplements are distinct.

For `u in B`, define

```
rho_u = residual degree into A,
q_u   = selected outdegree,
p_u   = selected indegree as supplement/exception.
```

For `i in A`, define

```
R_i = residual degree into B,
x_i = selected degree into B.
```

Let

```
r=sum_u rho_u=sum_i R_i.
```

3. Exact ledger and label demand

Selected cross-edges and existing edges of `H[B]` partition the unordered pairs of B, so

```
#selected + e(H[B]) = C(b,2).
```

Direct edge counting gives

```
e(F)=r+t,                                         (3.1)
sum_i d_i=2(r+t).                                 (3.2)
```

Minimum degree in H gives `d_H(i)>=a` for every `i in A`. Since

```
d_H(i)=1+(a-1-d_i)+R_i+x_i,
```

we have

```
x_i>=d_i-R_i.
```

Define

```
s_i=max(0,d_i-R_i),
S=sum_i s_i.
```

Then

```
x_i>=s_i.                                         (3.3)
```

Also

```
S >= sum_i(d_i-R_i)
  = 2(r+t)-r
  = r+2t.                                         (3.4)
```

Thus a label of demand `s_i` genuinely requires at least `s_i` distinct selected source incidences.

4. Pointwise forcing from a selected edge

Fix a selected edge `ui->w`.

4.1 `d_i<=rho_u+R_i`

Let j be an F-neighbour of i, so `ij` is absent from `H[A]`. Since `{u,i}` must dominate j, `uj` is an H-edge.

If `uj` is residual, charge j to a residual source slot at u. Otherwise `uj` is selected, say `uj->z`. Distinct selected labels at u have distinct supplements, so `z!=w`. Since `ui->w` must dominate z and `uz` is missing in `H[B]`, `iz` is an H-edge.

Moreover `iz` cannot be selected: i and z both miss the A-vertex j (`ij` is an F-edge and j is the unique exception of `uj->z`). A selected A-B edge has its unique exception in B and therefore must dominate every A-vertex. Hence `iz` is residual.

Distinct selected j give distinct supplements z, so the non-residual `uj` cases inject into distinct residual edges at label i. Therefore

```
d_i<=rho_u+R_i.                              (4.1)
```

In particular, at every selected incidence,

```
s_i<=rho_u.                                  (4.2)
```

4.2 `d_i<=rho_u+rho_w`

For the same F-neighbour j, if `uj` is residual charge it to rho_u. Otherwise write `uj->z`. Since `{u,j}` must dominate w and `uw` is missing, `jw` is an H-edge.

The edge `jw` cannot be selected from source w: j and w both miss the A-vertex i (`ij` is an F-edge and `iw` is missing because w is the exception of `ui->w`). Hence `jw` is residual. Distinct j give distinct residual edges at w. Therefore

```
d_i<=rho_u+rho_w.                            (4.3)
```

4.3 Source degree and supplement forcing

Every F-neighbour j of i is a cross-neighbour of u. Source u has exactly `rho_u+q_u` cross-neighbours, one of which is i, so

```
d_i<=rho_u+q_u-1.                          (4.4)
```

For every other selected `uj->z` at u, the pair `{u,j}` must dominate w. Since `uw` is missing, `jw` is an H-edge. The `q_u-1` other selected labels are distinct, so w has at least `q_u-1` cross-neighbours among them. These are partitioned into residual and selected edges from w, hence

```
rho_w+q_w>=q_u-1.                          (4.5)
```

5. Missing degree in B and endpoint load

Every missing unordered B-pair incident with u is oriented exactly once, either outward from u or inward to u. Hence

```
q_u+p_u = missing degree of u in H[B].     (5.1)
```

Since

```
d_H(u)=rho_u+(b-1)-(q_u+p_u)>=a,
```

we obtain

```
p_u<=rho_u+(b-a-1)=rho_u+3.                 (5.2)
```

Also

```
q_u+p_u<=15,
q_u+rho_u<=12.                              (5.3)
```

For selected `ui->w`, label i is adjacent in B to:

- u itself;
- the `q_u-1` supplements of the other outward selected pairs at u;
- the `p_u` sources of selected pairs oriented into u.

These vertices are distinct. An incoming source cannot equal an outgoing supplement because that would orient the same missing unordered B-pair both ways. Therefore

```
R_i+x_i>=q_u+p_u.                           (5.4)
```

6. Residual activity for positive surplus

Assume `t>0`. We prove

```
rho_u>=1 for every u in B.                  (6.1)
```

Suppose instead `rho_u=0`. Put

```
U=N_A(u),
T=A\U.
```

Every cross-edge from u is then selected. There is no F-edge between U and T: if `i in U`, `j in T` and `ij in F`, the selected edge at `ui` would have to dominate j, forcing `uj in H`, contradiction.

Now count forced residual edges.

For every F-edge `ij` inside U, the selected edges from u to i and j have distinct supplements. Cross-domination forces two residual edges, one at each A-endpoint. They are residual because each forced edge has endpoints that jointly miss an A-vertex. The ordered F-edge endpoints inject into these residual cross-edges, giving `2e(F[U])` distinct residual edges with A-endpoint in U.

Now take an F-edge `ij` inside T. Adding ij to H creates a quasi-edge with exception i or j. Its auxiliary cannot be v, because the pair with v already dominates the opposite A-endpoint; it cannot be u because u is adjacent to neither endpoint. If its auxiliary z lay in A, domination of u would force `z in U`. But a quasi-edge with exception in T would then require an absent U-T pair, contradicting `F(U,T)=empty`. Hence the auxiliary lies in B.

That cross quasi-edge has its unique exception in A, so it cannot be one of the selected representatives whose exception lies in B; it is residual. Its A-endpoint together with the unique exception recovers the original F-edge, so distinct F[T]-edges give distinct residual cross-edges. Their A-endpoints lie in T and are therefore disjoint from the U-family.

Thus

```
r>=2e(F[U])+e(F[T])
 >=e(F[U])+e(F[T])
 =e(F)
 =r+t,
```

contradicting `t>0`. Therefore (6.1) holds, and in particular

```
r>=b=16.                                    (6.2)
```

7. Charging inequality

For each label i choose exactly `s_i` of its actual selected incidences, possible by (3.3). By (4.2), a chosen source u for label i satisfies

```
rho_u>=s_i.
```

Also `q_u<=a-rho_u`. Charge each chosen incidence from source u by

```
(rho_u-1)/(a-rho_u).
```

A chosen source has `q_u>=1`, so `rho_u<=a-1` and the denominator is positive. Source u participates in at most `q_u<=a-rho_u` chosen incidences, so its total charge is at most `rho_u-1`. Summing over all B-sources gives total charge at most

```
r-b.
```

The charge function is increasing in integer rho on `1<=rho<a`. A label with demand `s_i` therefore receives at least

```
s_i(s_i-1)/(a-s_i).
```

Hence

```
r-b >= sum_i s_i(s_i-1)/(a-s_i).           (7.1)
```

Combining (7.1) with `S>=r+2t` gives

```
sum_i s_i(a+1-2s_i)/(a-s_i) >= b+2t.      (7.2)
```

At `a=12,b=16`,

```
sum_i s_i(13-2s_i)/(12-s_i) >= 16+2t.    (7.3)
```

8. Threshold-capacity lemma — complete proof

Fix an integer `h>=1` and define

```
I_h={i:s_i>=h},
W_h=sum_{i in I_h}s_i,
Z_h={u:rho_u>=h},
z_h=|Z_h|.
```

Choose `s_i` actual selected incidences for each `i in I_h`. By (4.2), every chosen source belongs to `Z_h`.

Call a selected incidence **heavy** if its label lies in `I_h`. Let `ell_u` be the number of actual heavy selected incidences from source u. Then

```
W_h<=sum_{u in Z_h} ell_u.                    (8.1)
```

Let

```
J={u in Z_h:ell_u>h},
j=|J|.
```

Fix `u in J` and a heavy selected edge `ui->w`. For each other heavy selected label k at u, `{u,k}` must dominate w; because `uw` is missing, `kw` is an H-edge. There are `ell_u-1>=h` such distinct heavy labels k.

We claim `w in Z_h`. If none of those forced edges `kw` is selected from source w, then all are residual and `rho_w>=ell_u-1>=h`. If at least one forced edge `kw` is selected from w, then its label k is heavy, so `s_k>=h`; applying (4.2) to that selected incidence gives `rho_w>=s_k>=h`. Thus every supplement of a heavy arc from J lies in `Z_h`.

Therefore every heavy selected arc contributed by a source in J uses an unordered B-pair entirely inside `Z_h` and incident with J. Because selection is injective on unordered B-pairs, the number of such arcs is at most

```
j(z_h-j)+C(j,2)=j*z_h-j(j+1)/2.             (8.2)
```

Every source in `Z_h\J` contributes at most h heavy arcs by definition of J. Combining with (8.1),

```
W_h <= (z_h-j)h + j*z_h - j(j+1)/2.         (8.3)
```

If `W_h>0`, some label has demand at least h and therefore needs at least h distinct sources in `Z_h`; so `z_h>=h`. Put

```
q=z_h-h.
```

Now

```
[h*z_h+C(q,2)] - RHS(8.3)
  = (q-j)(q-j-1)/2
  >=0,                                        (8.4)
```

because `q-j` is an integer and the product of two consecutive integers is nonnegative. Thus

```
W_h<=h*z_h+C(z_h-h,2),
```

or equivalently

```
2W_h<=z_h^2-z_h+h(h+1).                      (8.5)
```

This is a necessary capacity bound only; no sufficiency is claimed.

Two useful consequences are immediate. If `H0=max_i s_i>0`, then

```
z_H0>=H0.                                     (8.6)
```

Also, residual activity gives every B-source baseline degree at least one, so for `h>=2`,

```
r>=b+z_h(h-1).                                (8.7)
```

Hence any upper bound `r<=rmax` implies

```
z_h<=floor((rmax-b)/(h-1)).                   (8.8)
```

9. Excluding an isolated vertex of C

Suppose x is isolated in `C=H[A]`. Put `X=A\{x}`.

Because x is C-isolated,

```
e(C)=e(C[X]).
```

From the exact ledger `e(C)+r=C(a,2)-t`, the number `P0` of missing H-edges inside X is

```
P0=C(a-1,2)-e(C[X])
   =r-(a-1-t).                          (9.1)
```

Take a missing pair ij inside X. Adding ij to H creates a new adjacent total-dominating pair using i or j. It cannot be `{i,j}` because both miss x. Suppose it is `iz->j`. The auxiliary z cannot be v: v already neighbours j, so adding ij does not newly make `{i,v}` total dominating. It cannot lie in A: every A-vertex other than x misses x or, if z=x, `ix` is not an edge. Hence `z in B`.

Thus every missing X-pair gives a cross quasi-edge with auxiliary in B and exception in A. Such an edge cannot be selected, because selected representatives have their unique exception in B. Hence it is residual. Distinct missing X-pairs give distinct residual cross-edges because a cross quasi-edge has a unique exception. Let P be this family. Then

```
|P|=P0=r-(a-1-t).                       (9.2)
```

Let Z be the set of B-endpoints used by P. For `z in Z`, choose `iz->j` from P. The pair `{i,z}` must dominate x, and i misses x, so `xz` is an H-edge. This edge is residual: if xz were selected for a missing B-pair, its selected pair would have to dominate every A-vertex, but `iz->j` implies z misses j while x is isolated in C and also misses j. Thus xz is residual. It lies outside P because its A-endpoint is x.

For each `z in B\Z`, residual activity supplies a residual edge incident with z; it is outside P because no edge of P has B-endpoint z. Therefore every one of the b vertices of B supplies a distinct residual edge outside P. Hence

```
r>=|P|+b.
```

Using (9.2),

```
b<=a-1-t.                               (9.3)
```

At `a=12,b=16`, (9.3) is impossible for every positive t of interest. Therefore

```
delta(C)>=1.                            (9.4)
```

Consequently

```
d_i<=10,
e(C)>=6,
r<=C(12,2)-t-6=60-t.                  (9.5)
```

10. The exact bridge passed to the finite kernel

Every actual n=29, Delta=16 graph in a positive-surplus scope induces data satisfying:

```
a=12, b=16;
1<=rho_u<=12;
0<=d_i<=10;
r=sum rho=sum R;
e(F)=r+t;
s_i=max(0,d_i-R_i);
S>=r+2t;
q_u+rho_u<=12;
p_u<=rho_u+3;
q_u+p_u<=15;
for selected ui->w:
  d_i<=rho_u+R_i,
  d_i<=rho_u+rho_w,
  d_i<=rho_u+q_u-1,
  rho_w+q_w>=q_u-1,
  R_i+x_i>=q_u+p_u;
charging inequality (7.3);
threshold-capacity inequalities (8.5)-(8.8);
r<=60-t.
```

The finite kernel deliberately enumerates a **superset** of graph-realizable arithmetic states satisfying necessary consequences of this bridge. Eliminating the relaxation is therefore safe provided every pruning rule is necessary and every terminal contradiction is checked exactly.

11. Hand cap for the upper range

For every integer `0<=s<=11`,

```
s(13-2s)/(12-s)<=5/2,                    (11.1)
```

because

```
5/2-s(13-2s)/(12-s)
 =(s-4)(4s-15)/(2(12-s))>=0.             (11.2)
```

Equality occurs only at `s=4`. With twelve labels, (7.3) gives

```
16+2t<=30,
```

so

```
t<=7,
m<=215.                                  (11.3)
```

Thus no Delta=16 graph exists at `m>=216`.

At `m=215`, `t=7`, equality in (11.3) forces every demand to equal four. Hence `S=48`. Equation (7.1) gives

```
r>=16+12*(4*3/8)=34,
```

while `S>=r+14` gives `r<=34`, so `r=34`. From residual activity,

```
34>=16+3z_4,
```

hence `z_4<=6`. But threshold capacity gives

`96=2W_4<=z_4^2-z_4+20<=50,`

contradiction. Therefore `m=215` is also excluded by hand.

The only upper-range Delta=16 scopes above 211 that still require finite arithmetic are consequently

`m=212,213,214.`

12. Review priorities

The highest-value independent attacks remain:

1. the quasi-edge construction and one-representative-per-unordered-pair convention;
2. the residual/non-selected status of the forced cross-edges in Section 4;
3. residual activity in Section 6;
4. the charging budget in Section 7;
5. threshold capacity in Section 8;
6. the isolated-C injection in Section 9;
7. the graph-to-averaged-LP embedding used after this bridge;
8. exact certificate semantics.

A single counterexample to a universal bridge lemma overrides every green computation downstream. This remains candidate mathematics until independent expert review is completed.